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(54) **SYSTEMS AND METHODS FOR DYNAMIC
SORTATION OF OBJECTS**

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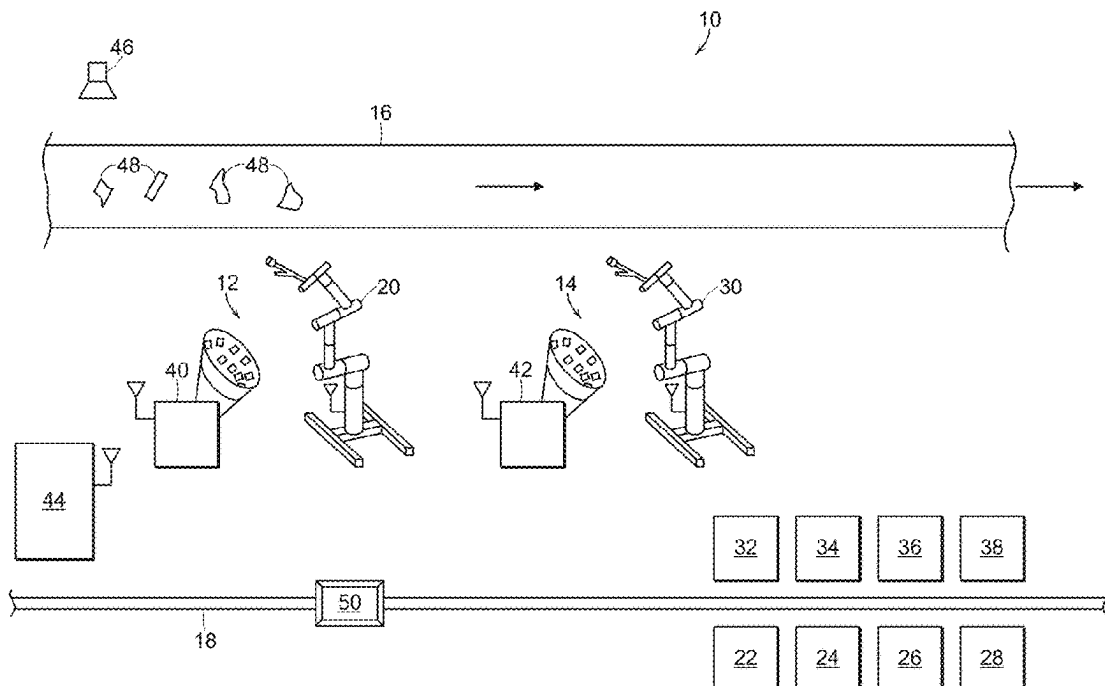
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continuation of application No. 17/828,314, filed on
May 31, 2022, now Pat. No. 11,839,902, which is a
continuation of application No. 16/902,351, filed on
Jun. 16, 2020, now Pat. No. 11,400,491, which is a

(57)

ABSTRACT

An automated shuttle sorter is disclosed that includes a
carriage that is movable from a load position at which the
carriage may be loaded, and at least two destination loca-
tions into which any contents of the carriage may be
provided from the carriage.



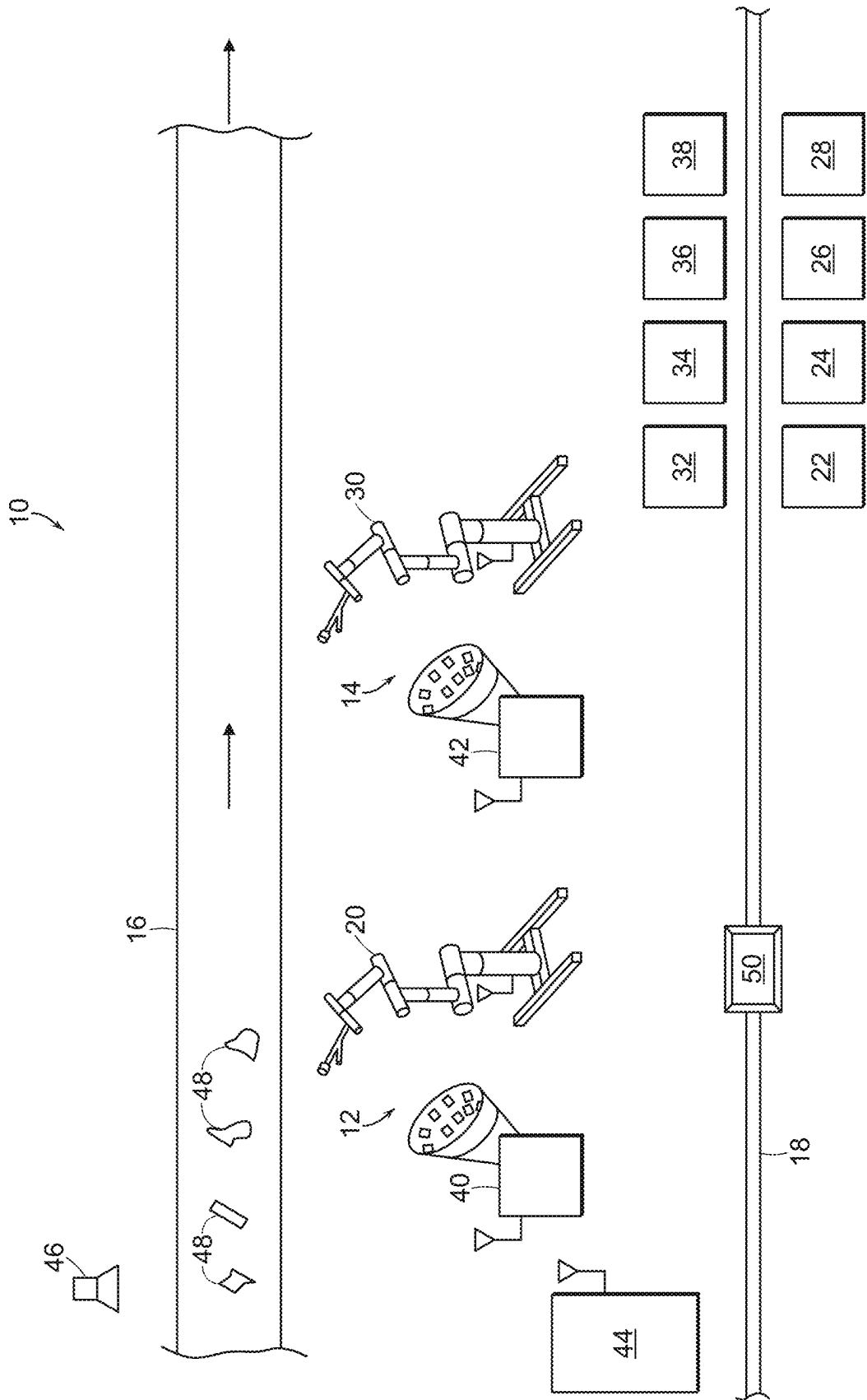


FIG. 1

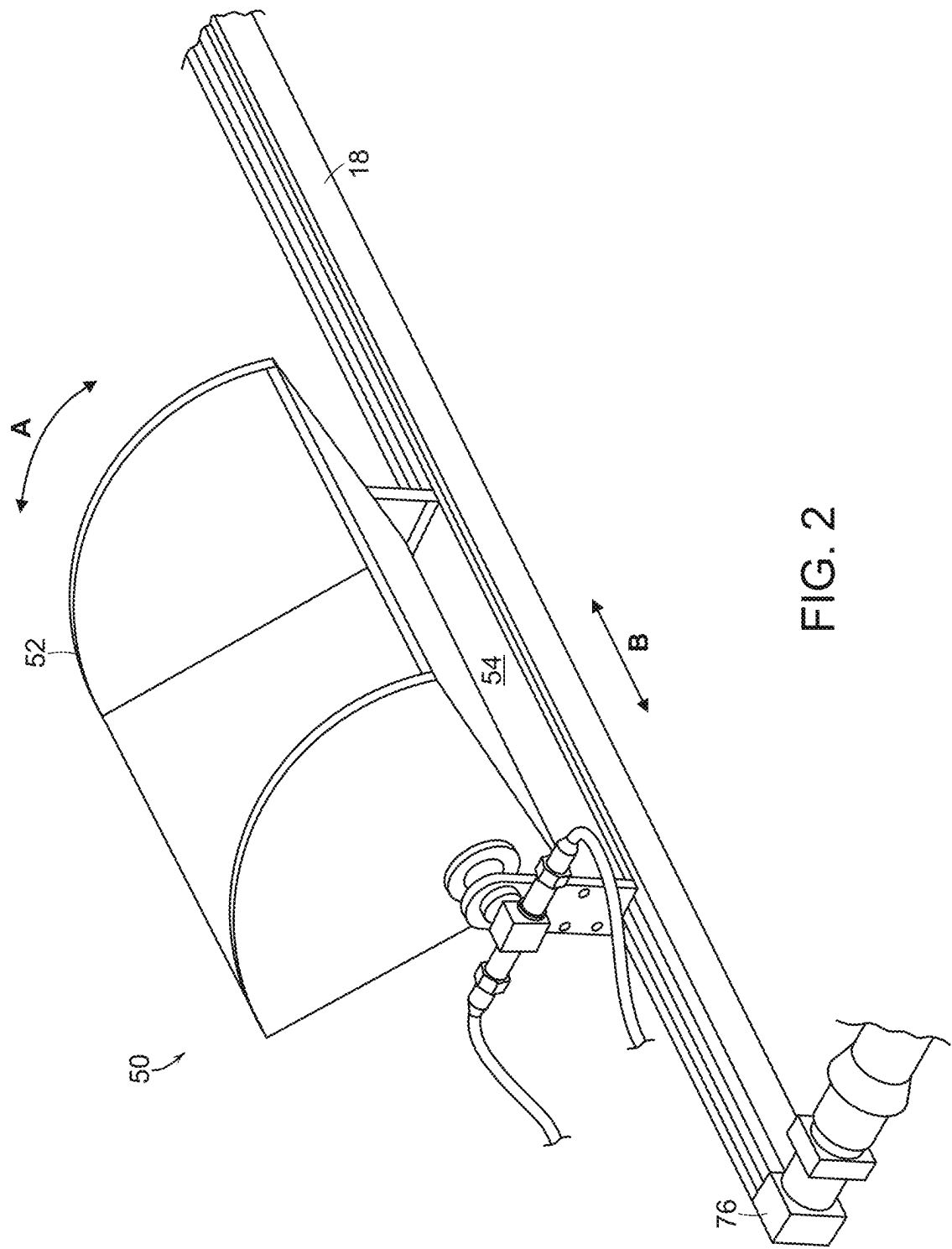
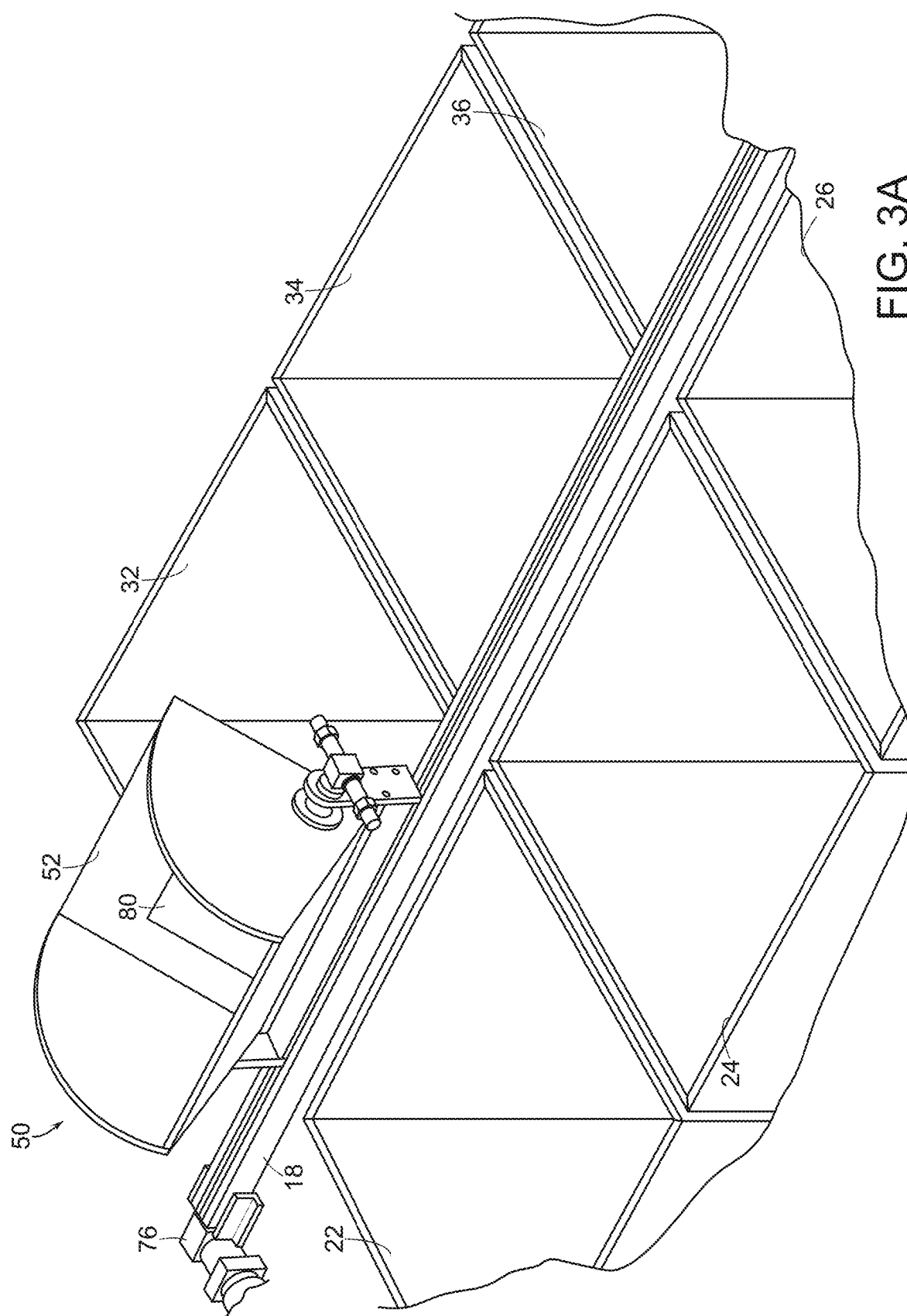


FIG. 2



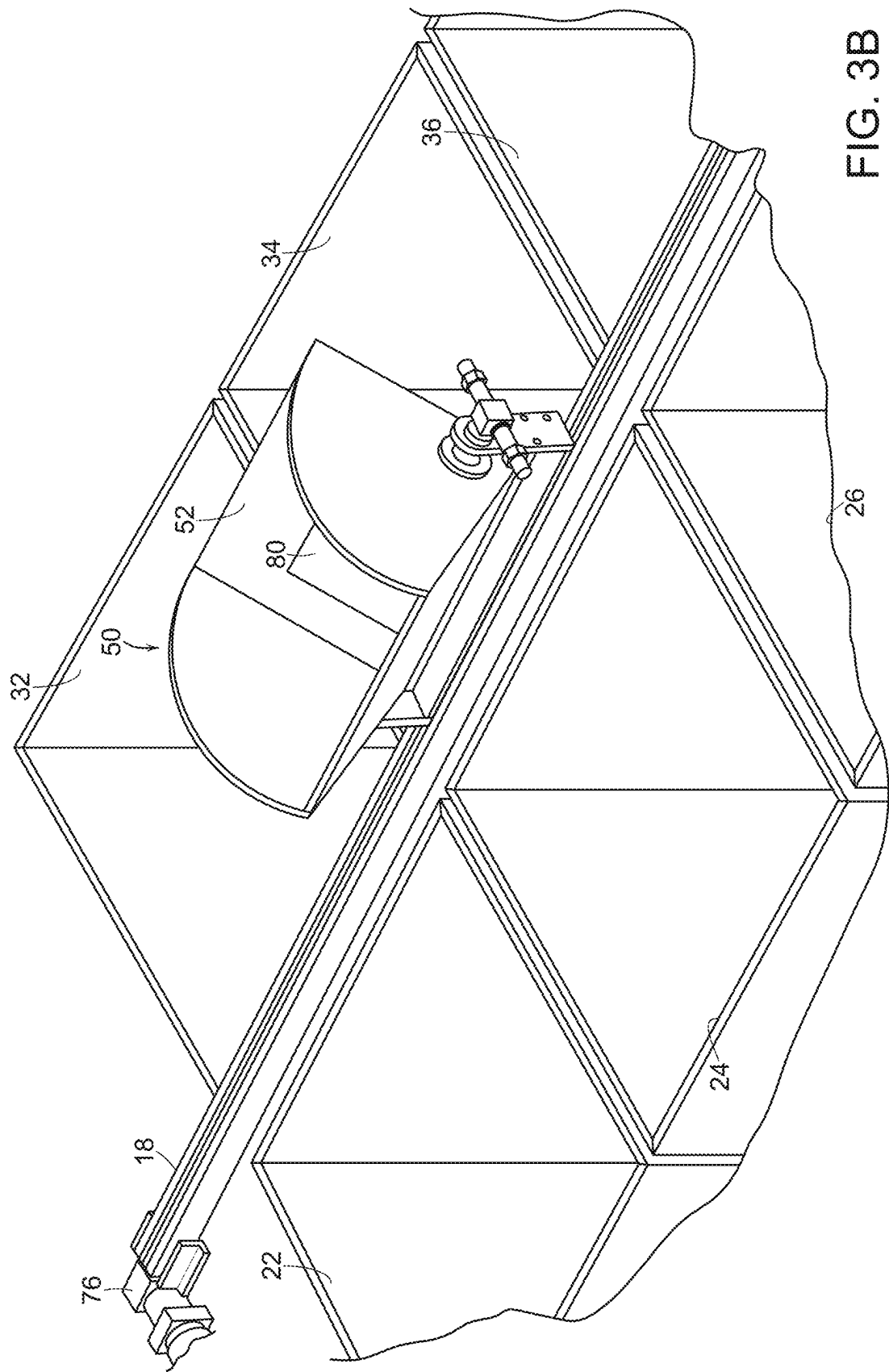


FIG. 3B

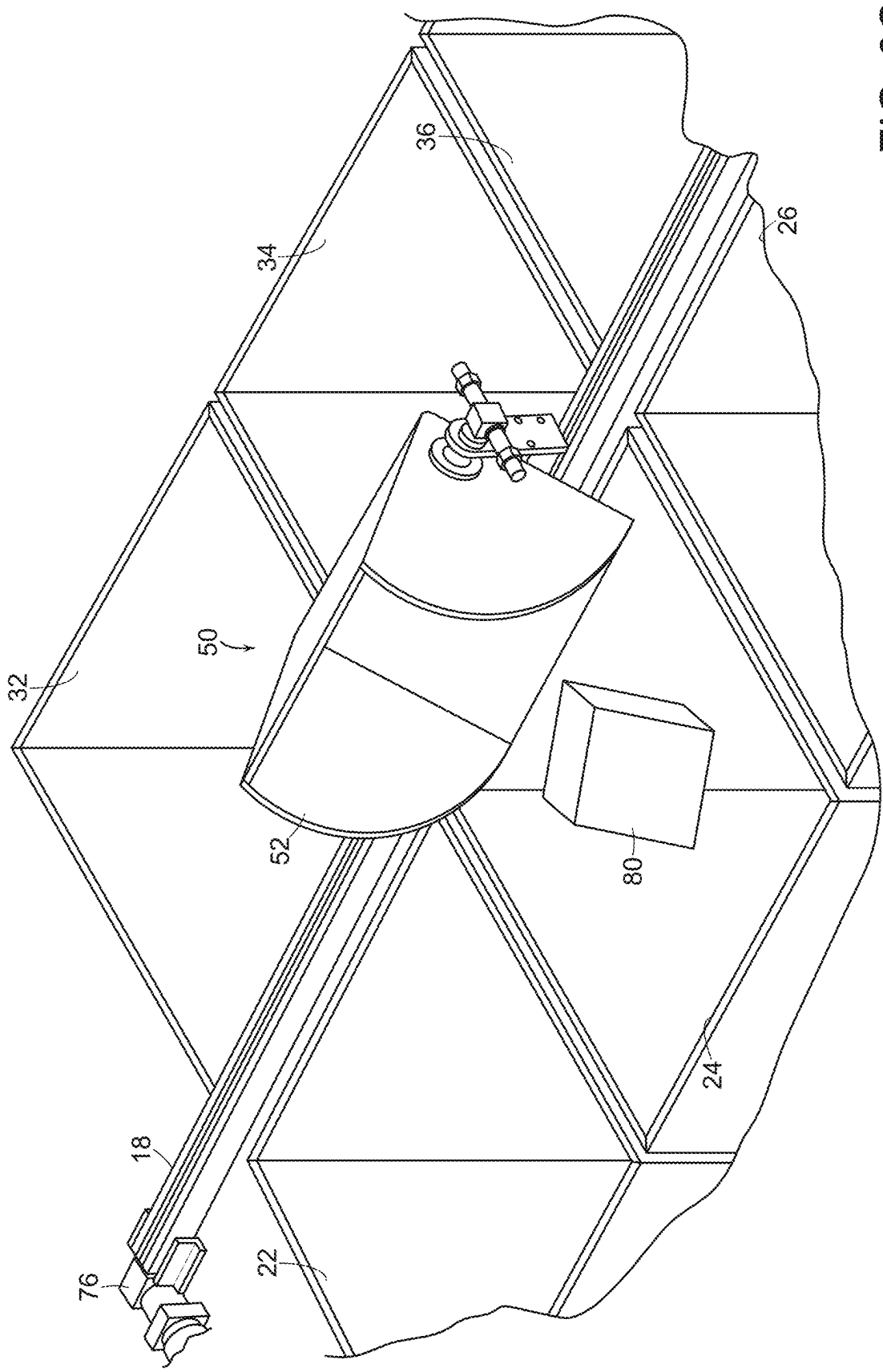


FIG. 3C

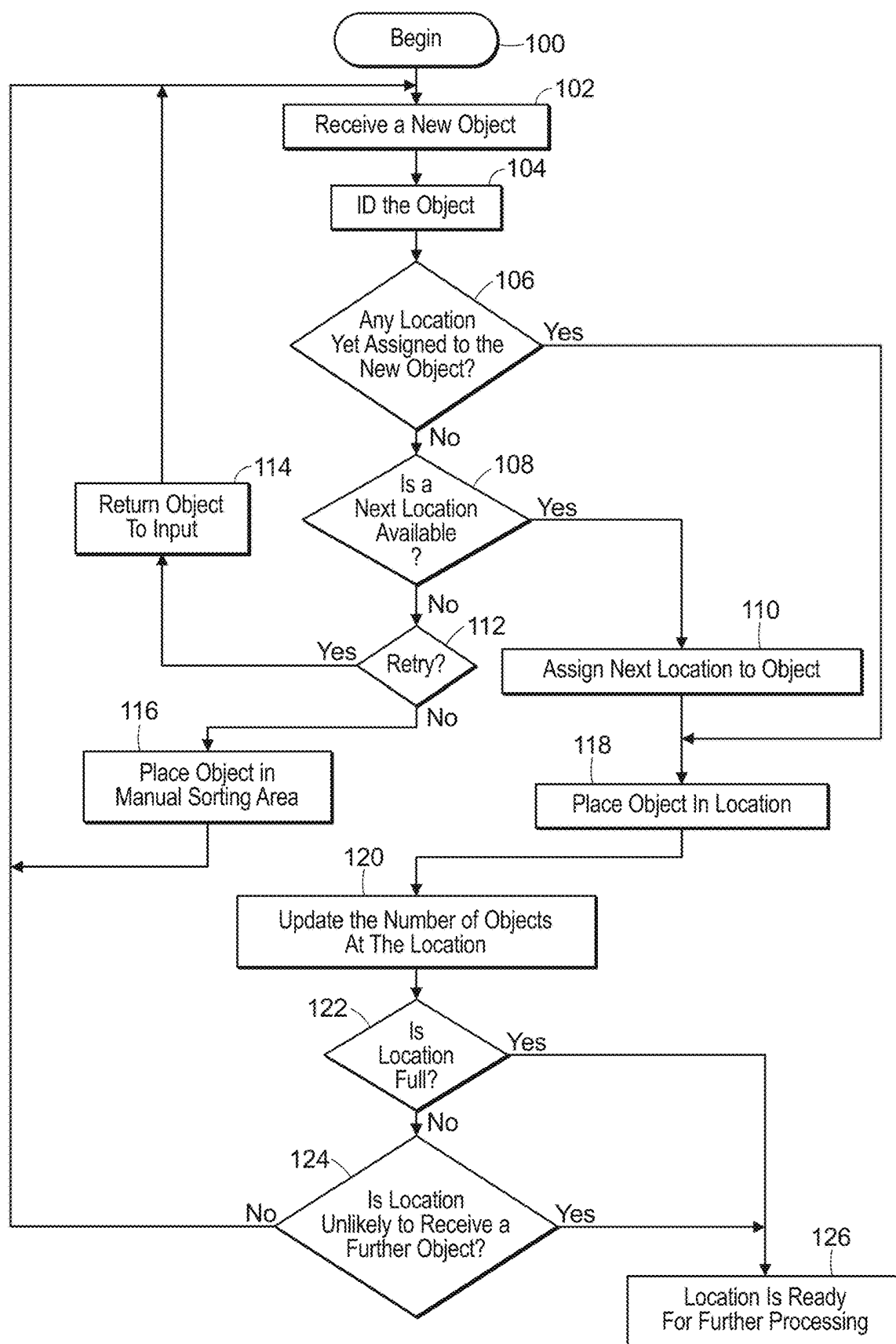


FIG. 4

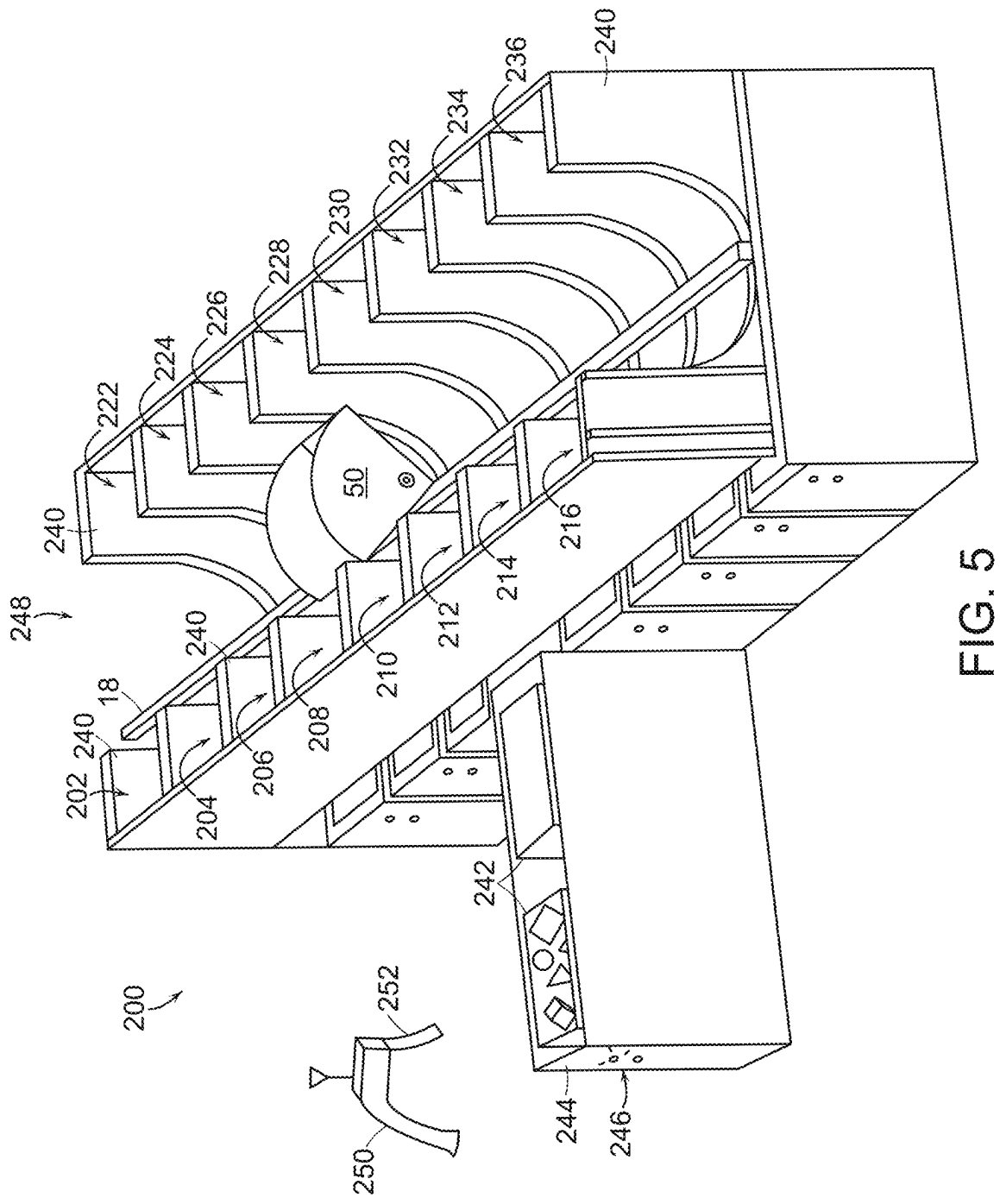


FIG. 5

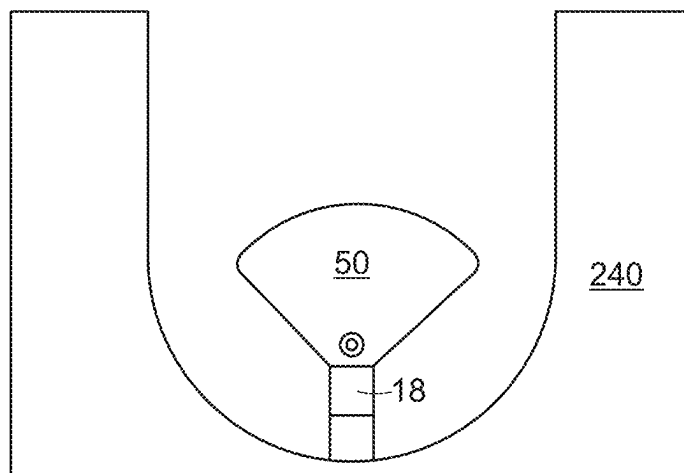


FIG. 6

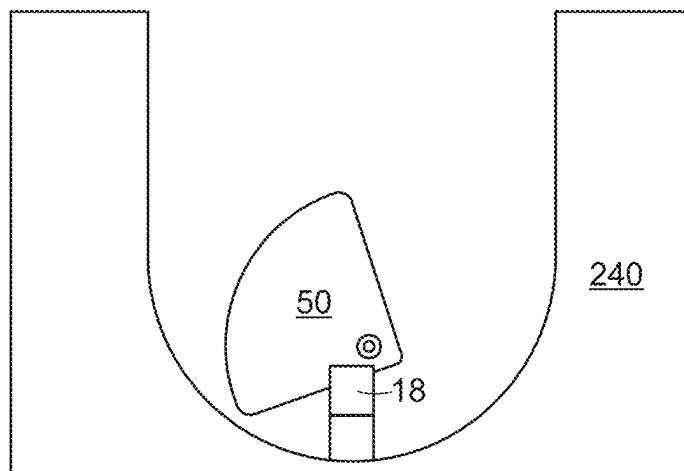
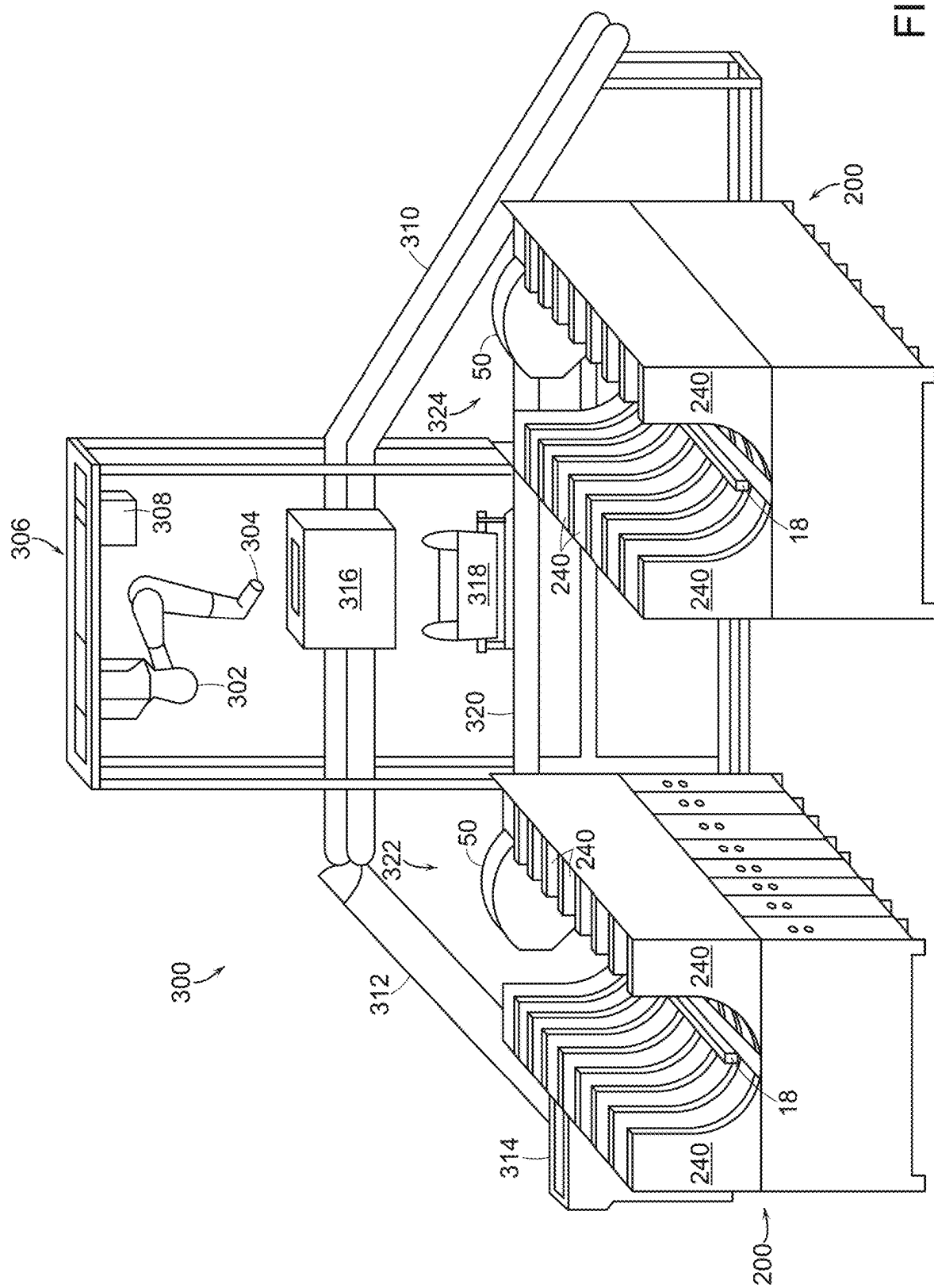
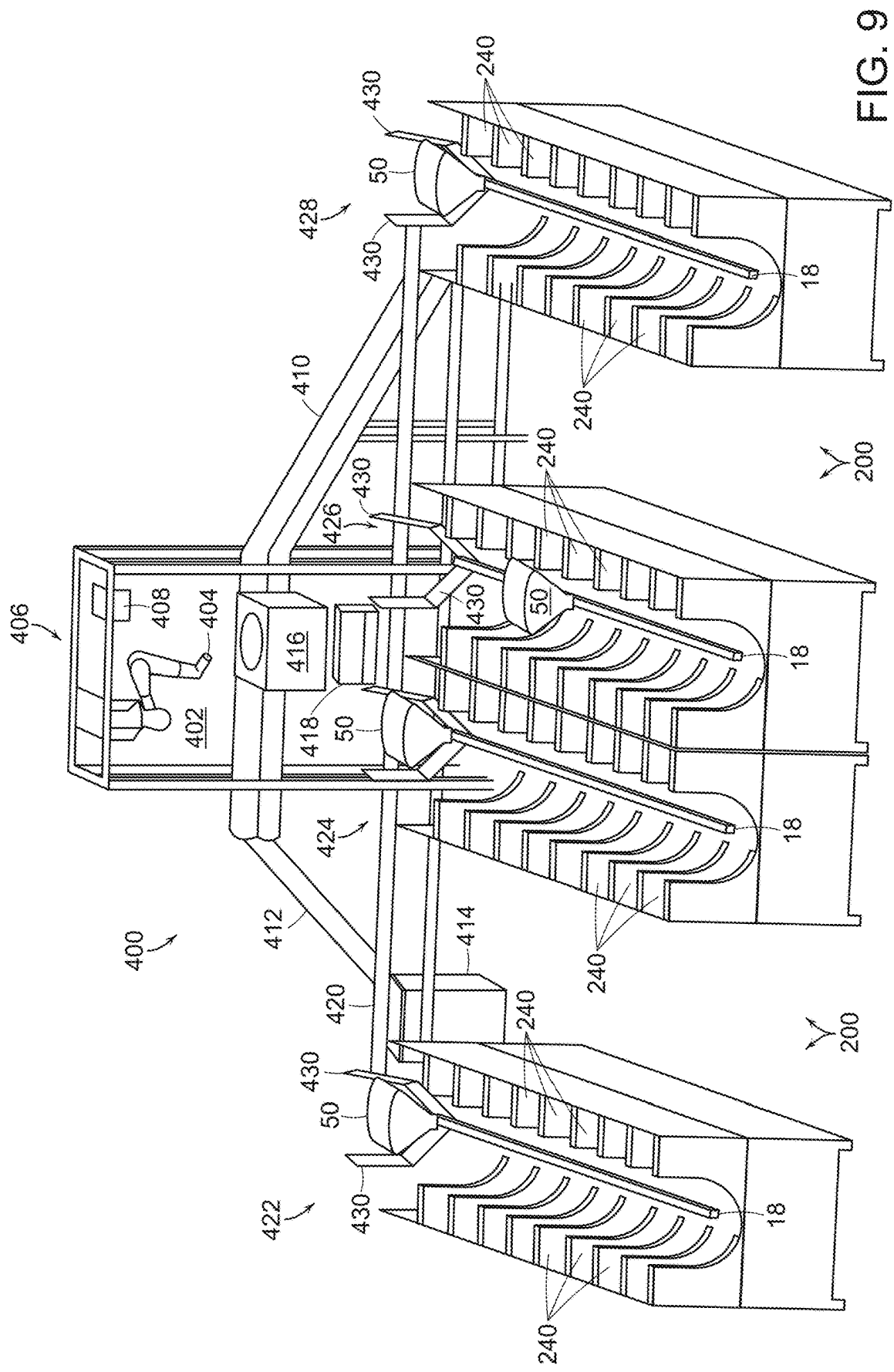


FIG. 7





 DEPARTMENT OF HEALTH AND HUMAN SERVICES



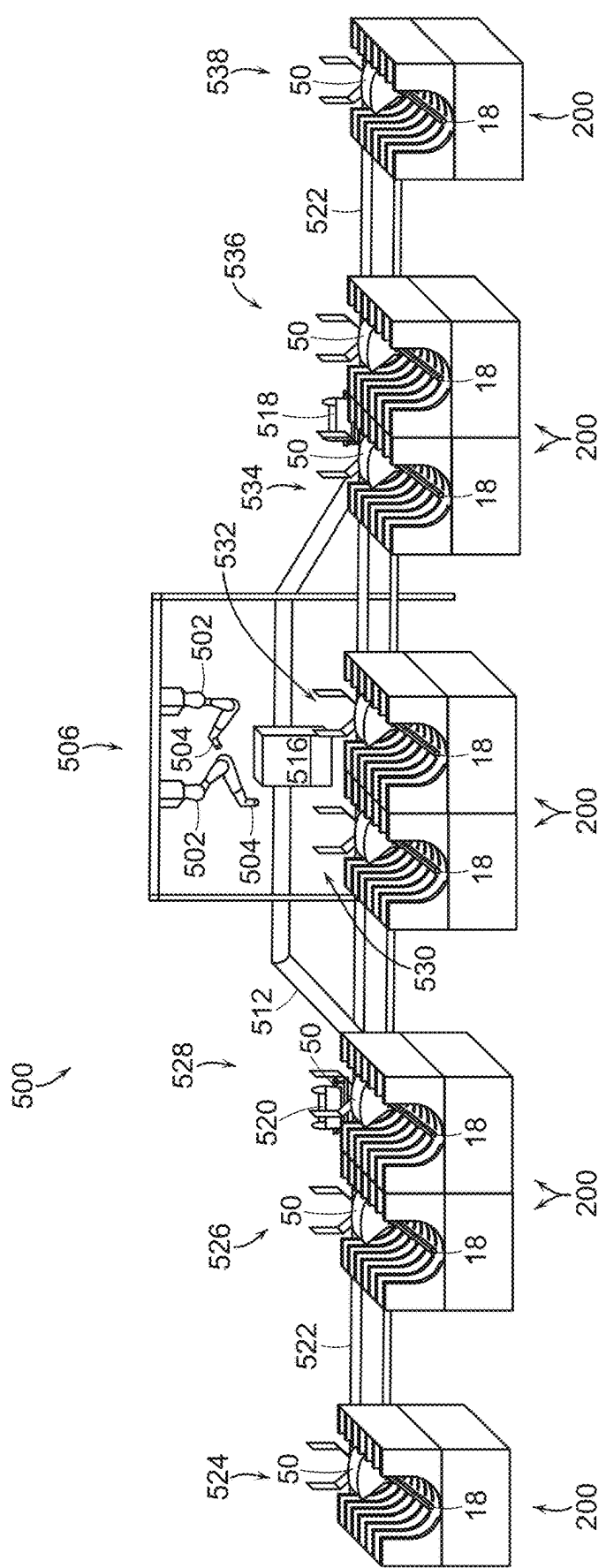


FIG. 10

SYSTEMS AND METHODS FOR DYNAMIC SORTATION OF OBJECTS

PRIORITY

[0001] The present application is a continuation of U.S. patent application Ser. No. 18/377,417, filed Oct. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/828,314, filed May 31, 2022, now U.S. Pat. No. 11,839,902, issued Dec. 12, 2023, which is a continuation of U.S. patent application Ser. No. 16/902,351, filed Jun. 16, 2020, now U.S. Pat. No. 11,400,491, issued Aug. 2, 2022, which is a continuation of U.S. patent application Ser. No. 15/241,779, filed Aug. 19, 2016, now U.S. Pat. No. 10,730,078 issued Aug. 4, 2020, which claims priority to U.S. Provisional Patent Application Ser. No. 62/263,050, filed Dec. 4, 2015, as well as U.S. Provisional Patent Application Ser. No. 62/265,181, filed Dec. 9, 2015, the entire disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND

[0002] The invention generally relates to sortation systems, and relates in particular to robotic and other sortation systems for sorting objects, where the sortation systems are intended to be used in dynamic environments requiring the systems to accommodate the processing of a variety of objects.

[0003] Current distribution center sorting systems, for example, generally assume an inflexible sequence of operations whereby a disorganized stream of input objects is first singulated into a single stream of isolated objects presented one at a time to a scanner that identifies the object. An induction element (e.g., a conveyor, a tilt tray, or manually movable bins) transport the objects to the desired destination or further processing station, which may be a bin, a chute, a bag or a conveyor etc.

[0004] In typical parcel sortation systems, human workers or automated systems typically retrieve parcels in an arrival order, and sort each parcel or object into a collection bin based on a set of given heuristics. For instance, all objects of like type might go to a collection bin, or all objects in a single customer order, or all objects destined for the same shipping destination, etc. The human workers or automated systems are required to receive objects and to move each to their assigned collection bin. If the number of different types of input (received) objects is large, a large number of collection bins is required.

[0005] Such a system has inherent inefficiencies as well as inflexibilities since the desired goal is to match incoming objects to assigned collection bins. Such systems may require a large number of collection bins (and therefore a large amount of physical space, large capital costs, and large operating costs) in part, because sorting all objects to all destinations at once is not always most efficient.

[0006] In short, when automating sortation of objects, there are a few main things to consider: 1) the overall system throughput (parcels sorted per hour), 2) the number of diverts (i.e., number of discrete locations to which an object can be routed), 3) the total area of sortation system (square feet), and 4) the annual costs to run the system (man-hours, electrical costs, cost of disposable components).

[0007] Current state-of-the-art sortation systems rely on human labor to some extent. Most solutions rely on a worker

that is performing sortation, by scanning an object from an induction area (chute, table, etc.) and placing the object in a staging location, conveyor, or collection bin. When a bin is full or the controlling software system decides that it needs to be emptied, another worker empties the bin into a bag, box, or other container, and sends that container on to the next processing step. Such a system has limits on throughput (i.e., how fast can human workers sort to or empty bins in this fashion) and on number of diverts (i.e., for a given bin size, only so many bins may be arranged to be within efficient reach of human workers).

[0008] Other partially automated sortation systems involve the use of recirculating conveyors and tilt trays, where the tilt trays receive objects by human sortation, and each tilt tray moves past a scanner. Each object is then scanned and moved to a pre-defined location assigned to the object. The tray then tilts to drop the object into the location. Further partially automated systems, such as the bomb-bay style recirculating conveyor, involve having trays open doors on the bottom of each tray at the time that the tray is positioned over a predefined chute, and the object is then dropped from the tray into the chute. Again, the objects are scanned while in the tray, which assumes that any identifying code is visible to the scanner.

[0009] Such partially automated systems are lacking in key areas. As noted, these conveyors have discrete trays that can be loaded with an object; they then pass through scan tunnels that scan the object and associate it with the tray in which it is riding. When the tray passes the correct bin, a trigger mechanism causes the tray to dump the object into the bin. A drawback with such systems however, is that every divert requires an actuator, which increases the mechanical complexity and the cost per divert can be very high.

[0010] An alternative is to use human labor to increase the number of diverts, or collection bins, available in the system. This decreases system installation costs, but increases the operating costs. Multiple cells may then work in parallel, effectively multiplying throughput linearly while keeping the number of expensive automated diverts at a minimum. Such diverts do not ID an object and cannot divert it to a particular spot, but rather they work with beam breaks or other sensors to seek to ensure that indiscriminate bunches of objects get appropriately diverted. The lower cost of such diverts coupled with the low number of diverts keep the overall system divert cost low.

[0011] Unfortunately, these systems don't address the limitations to total number of system bins. The system is simply diverting an equal share of the total objects to each parallel manual cell. Thus each parallel sortation cell must have all the same collection bins designations; otherwise an object might be delivered to a cell that does not have a bin to which that object is mapped.

[0012] There remains a need for a more efficient and more cost effective object sortation system that sorts objects of a variety of sizes and weights into appropriate collection bins or trays of fixed sizes, yet is efficient in handling objects of such varying sizes and weights.

SUMMARY

[0013] In accordance with an embodiment, the invention provides an automated shuttle sorter that includes a carriage that is movable from a load position at which the carriage

may be loaded, and at least two destination locations into which any contents of the carriage may be provided from the carriage.

[0014] In accordance with another embodiment, the invention provides a sortation system that includes an automated carriage for receiving an object at a load station from an object identification system. The automated carriage includes an automated transport system for reciprocally moving between at least two destination stations, and a transfer system for transferring the object from the automated carriage into one of the at least two destination stations.

[0015] In accordance with a further embodiment, the invention provides a method of sorting objects. The method includes the steps of acquiring an object to be sorted from an input station, identifying the object, providing the object to an automated carriage that is reciprocally movable between at least two destination stations, and moving the object to one of the at least two destination stations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The following description may be further understood with reference to the accompanying drawings in which:

[0017] FIG. 1 shows an illustrative diagrammatic view of a sortation system in accordance with an embodiment of the invention;

[0018] FIG. 2 shows an illustrative diagrammatic view of a carriage for use in the sortation system of FIG. 1;

[0019] FIGS. 3A-3C show illustrative diagrammatic views of the carriage of FIG. 2 at different stages of movement;

[0020] FIG. 4 shows an illustrative flowchart showing processing steps during operation of a system in accordance with an embodiment of the invention;

[0021] FIG. 5 shows an illustrative diagrammatic view of a sortation system in accordance with another embodiment of the invention;

[0022] FIGS. 6 and 7 show illustrative diagrammatic views of a portion of the system of FIG. 5;

[0023] FIG. 8 shows an illustrative diagrammatic view of a sortation system in accordance with a further embodiment of the invention that includes two carriages;

[0024] FIG. 9 shows an illustrative diagrammatic view of a sortation system in accordance with a further embodiment of the invention that includes four carriages; and

[0025] FIG. 10 shows an illustrative diagrammatic view of a sortation system in accordance with a further embodiment of the invention that includes eight carriages.

[0026] The drawings are shown for illustrative purposes only.

DETAILED DESCRIPTION

[0027] In accordance with various embodiments, the invention provides an inherently more flexible object sortation system in which objects may be selected in a most advantageous order, and the sortation of those objects may take advantage of dynamically varying correspondence between the sorter outputs and the ultimate object destinations. The invention further provides a highly efficient and readily scalable system for providing the transport and distribution of objects.

[0028] Systems and methods of the present invention are well suited to applications in current sortation systems that receive objects in a disorganized stream and are required to sort the objects into sorted streams. Such systems recognize that reading information on an object may sometimes be challenging, so that once an object is scanned, it is important to keep the information associated with the object. The acquisition of objects from disorganized jumbles is challenging, and once an object is acquired, it is important to keep the object separated from other objects. Further, the transport and conveying systems have limited flexibility, typically following a single track that passes every possible destination.

[0029] In accordance with certain embodiments, the invention provides systems and methods that upend basic assumptions of current sortation systems, with improvements in each of the challenges identified above. The systems, in some embodiments, provide improved transport and distribution, and further provide for the identification of the entire object's shape and disposition, reducing or eliminating the need to keep the object separate from others. The use of robotic manipulators improves the reliability and economy of acquiring objects, even when in a jumble with other objects, reducing the need to maintain separation of objects. The systems, in further embodiments, provide improved transport and conveyor systems, and provide learning algorithms in particular, that allow dynamically changing patterns of object handling, with resulting efficiencies in the sortation process, lower space requirements, lower demand for manual operations, and as a consequence, lower capital and operating costs for the entire system.

[0030] FIG. 1, for example, shows a system 10 in accordance with an embodiment of the present invention that includes a sorting station 12 that is fed by a common input conveyor 16. An output track 18 carries an output carriage 50 to bins 22, 24, 26, 28, 32, 34, 36 and 38. The carriage 50 brings items to a dynamically assigned bin. The sorting station 12 may also include a scanner 40 that permits an object to be scanned while being held by an end effector of a robotic system 20. A central controller 44 communicates with the robotic system 20 and scanner 40 to provide input regarding the assignment of objects to a bin as discussed in more detail below. An additional scanner 46 may be employed to provide the sorting stations with advance information regarding objects 48 that are being provided on the input conveyor 16. The system may also include further sorting stations, e.g., 14, that may also include a scanner 42 that permits an object to be scanned while being held by an end effector of a robotic system 30. A central controller 44 communicates with the robotic system 30 and scanner 42 to provide input regarding the assignment of objects to a bin as discussed in more detail below.

[0031] During use, each sorting station 12, 14 may either select an object and then identify the selected object by a detection device on the articulated arm, or may use the articulated arm to hold the object in front of a scanner, or may place the object into a scanner as discussed below, or may first identify an object prior to selection, and then grasp the identified object. In any event, the system then assigns a bin to the object if a new bin is available and the object is not yet assigned a bin at that sorting station. What is significant is that the sorting station is not pre-assigned a large set of collection bins assigned to all possible objects that may appear in the input path. Further, the central

controller may employ a wide variety of heuristics that may further shape the process of dynamically assigning objects to collection bins as discussed in more detail below. The input conveyor may be also provided as a loop conveyor on which objects pass by multiple sorting stations, or the input conveyor may be provided as multiple conveyors on which objects pass by multiple sorting stations. The invention provides, therefore, examples of sortation systems that involve moving infed objects directly to a buffer, without human intervention. The buffer holds the objects, possibly in a disorganized jumble, where they may be accessed by one of several sorters. The perception system may read labels when they are visible, but may also use more general machine vision algorithms to identify object class and shape, and to track objects as they are circulated. The sorters acquire objects from the buffer. If needed, they use their own perception systems to read labels not previously read. They may move objects to any of several outputs, including the possibility of placing an object back on the buffer, either for later handling or for handling by a different sorter.

[0032] Systems of various embodiments provide numerous advantages because of the inherent dynamic flexibility. The flexible correspondence between sorter outputs and destinations provides that there may be fewer sorter outputs than destinations, so the entire system may require less space. The flexible correspondence between sorter outputs and destinations also provides that the system may choose the most efficient order in which to handle objects, in a way that varies with the particular mix of objects and downstream demand. The system is also easily scalable, by adding sorters, and more robust since the failure of a single sorter might be handled dynamically without even stopping the system. It should be possible for sorters to exercise discretion in the order of objects, favoring objects that need to be handled quickly, or favoring objects for which the given sorter may have a specialized gripper.

[0033] The system may also employ a flexible destination stage, including a process for dynamically changing the correspondence of sorter outputs and system destinations using a switch based on heuristics from the sortation process. The system may dynamically map sorter outputs to system destinations based on long-term historical usage trends and statistics, or items already processed, or current contents of other dynamically allocated sorter outputs, or average, minimum or maximum time-to-sort associated with each sorter output, or physical characteristics of the items sorted, or a priori information, or known future deliveries, or location within a facility, including the physical location relative to other allocated sorter outputs (e.g., above, beside, on or nearby), or incoming shipments, as well as knowing what items are currently upstream of the sortation process and combinations of the above. Further, systems of embodiments of the invention provide that information regarding correspondence between sorter outputs to system destinations may be provided to an automated system for sorting.

[0034] By making use of heuristics, the mapping of sorter outputs to system destinations can be improved substantially over traditional fixed allocation. Destinations may be assigned on the fly, reducing wasted space from unused sorter outputs and decreasing the time it takes to process incoming objects. Long-term historic trends may be used to allocate sorter outputs when the next incoming group of objects is either in-part or entirely unknown. Historical usage patterns provide insight into when objects bound for

certain destinations can be expected to arrive, the number of objects bound for each destination expected for any given time, and the probable physical properties of these incoming objects.

[0035] In addition to trends pertaining to incoming objects, historical trends provide information on the speed at which objects can be sorted into outputs, and the rate at which outputs are transferred to system destinations. These factors allow sorter outputs to be allocated probabilistically until a deterministic understanding of incoming objects is achieved.

[0036] In addition to historic trends, an understanding of the current state of the system is used to ensure that there is an appropriate amount of space allocated for those objects that are expected to arrive. When combined with the knowledge of those objects that have already been sorted, the correspondence of sorter outputs to system destinations can typically be allocated deterministically. A knowledge of those objects already processed and the contents of current sorter outputs allows the system to optionally remap the sorter outputs once they have been emptied of their contents. In the case that there aren't enough sorter outputs, this knowledge also allows the system to specify which sorter outputs should be emptied such that they can quickly be reallocated to new system destinations.

[0037] A further consideration when dynamically allocating sorter outputs is to take into account the physical characteristics of the packages and the facility. If a certain destination is expected to receive larger, unwieldy objects, then an appropriately-sized sorter output can be allocated. If a particular system destination will require more than a single sorter output, then two adjacent outputs can be allocated with the same destination in order to facilitate human intervention.

[0038] A method is also presented for displaying the sorter output-system destination correspondence information next to the destinations. This allows human workers interacting with the system to understand how and when to properly empty the destinations. In addition, critical to autonomous sortation is the ability to send these destination allocations to a sortation system without human intervention. This allows for the construction of fully-streamlined sortation system software.

[0039] The carriage system for placement at destination locations is also efficient and scalable. As shown in FIG. 2, for example, the carriage **50** includes a load bed **52** that is pivotally mounted on a carrier **54** (as generally indicated at A). The carrier **54** is slidably mounted on the track **18** (as generally indicated at B). The movement of the carriage may be provided by any of a variety of power sources, such as electric charge via an electric track, or by pneumatics or a belt drive that drive the carriage along the track in both directions with good speed and accuracy. Similarly, the tipping of the carriage may be provided by any of a variety of power sources such as electric charge (e.g., using reverse direction solenoids), or by pneumatics that tip the carriage in either direction (transverse to the direction of the track) as desired.

[0040] For example, FIGS. 3A-3C show the carriage **50** conveying an object **80** within the load bed **52** of the carriage **50** from a first location (FIG. 3A) to a destination location (FIG. 3B), whereupon the load bed **52** is tipped (FIG. 3C),

causing the object **80** to fall into the a desired destination location **24** among a plurality of other destination locations **22, 26, 32, 34, 36**.

[0041] In accordance with certain embodiments therefore, systems of the invention may employ carriages that shuttle back and forth along shuttle directions. Such systems may rely on a pre-sortation step, where an object is sorted first to the correct sortation station, and once there, it is sorted into the proper collection bin. In this fashion, different stations can have different collection bin mappings, allowing the total number of system bins to be multiplied by the number of parallel sortation stations operating. Such pre-sortation steps however, must be either complicated and expensive automated systems, or must rely on yet more human work; either way adds cost which raises the overall cost per divert of the system to unacceptably high levels. The invention provides a new approach to object sortation that yields a large (and very flexible and scalable) number of total collection bins, very low divert costs per bin, throughput as high as that of a manual system, and a far smaller need for manual labor to operate.

[0042] FIG. 4 shows a flowchart of the operation of a system in accordance with an embodiment of the present invention. The process begins (step **100**) and the articulated arm, or another object reception device, receives a new object (step **102**). The system then identifies the new object (step **104**) by any of an overhead scanner **46**, or a scanner system **40, 42**, or by a drop scanner as discussed below, etc. The system then determines whether any location at the station has yet been assigned to the new object (step **106**). If so, the system then places the object at that location (step **118**). If not, the system then determines whether a next location is available (Step **108**). If not, the system may (either with or without input from a human) determine whether to retry identifying the object (step **112**). If so, then the system would return the object to the input stream (step **114**) to be again received at a later time (step **102**). If not, the system would place the object in a manual sorting area for sortation by a human (step **116**). If a next location is available (step **108**), the system then assigns a next location to the object (step **110**), and the object is then placed in that location (step **118**). If a location had already been assigned to the object (step **106**), then the system the object is placed in that location (step **118**). The number of objects at the location is then updated (step **120**), and if the location is then full (step **122**), the system identifies that the location is ready for further processing (step **126**). If not, the system then determines whether (based on prior knowledge and/or heuristics), whether the location is likely to receive a further object (step **124**). If so, the system identifies that the location is ready for further processing (step **126**). If not, the system returns to receiving a new object (step **102**). The further processing may, for example include collecting the items at the location in a single bag for transport to shipping to a geographic area.

[0043] FIG. 5 shows, for example, a shuttle system **200** that includes a carriage **50** on a track **18** that accesses locations **202, 204, 206, 208, 210, 212, 214, 216, 222, 224, 226, 228, 230, 232, 234** and **236**. Each location may include chute walls **240** for guiding the objects, and bins **242** for receiving the objects. Generally, objects are serially loaded into the carriage **50** at a first end **248**, and the carriage **50** shuttles the objects to the assigned bins. The bins **242** may be provided on drawers **244** that slide out from the system

200, wherein the bins for locations **212** and **232** are shown pulled out. Each drawer **244** may include lights **246** that become illuminated when the system identifies that a bin is ready for further processing. As shown in FIG. 9, the light associated with the bin at location **212** is shown lit because the bin has been identified as being full. A hand-held printer (or printer/scanner) **250** may be in wireless communication with the central controller **44**, and may print out a label **252** specifically identifying the contents of the bin that is being emptied for further processing. In certain embodiments, for example, the bin may include a bag that is sealed and labeled by a label **252** when pulled from the drawer. As shown in FIGS. 6 and 7, when the carriage is tipped on the track **18**, the carriage **18** does not enter into the area defined by the chute walls. This permits the carriage **18** to begin its travel back to the first end **248** as soon as the object is ejected from the carriage.

[0044] FIG. 8 shows a shuttle sortation system **300** that includes two shuttle systems **200** of FIG. 5, each of which includes a carriage **50** on a track **18** that provides objects to bins **242**. The system **300** also includes an articulated arm **302** with an end effector **304**, an input area **306** in which objects are presented for sortation, an overhead camera **308** for identifying objects to be sorted, and a receiving conveyor **310** for receiving objects to be sorted from any of a human worker, another conveyor, or an input pan. The system also includes a non-sortable output chute **312** that leads to a non-sortable output bin **314** for providing objects that the system either could not identify or could not sort for any other reason (e.g., could not grasp or pick up).

[0045] In addition to the overhead camera **308**, the system also includes a drop scanner **316** that includes an open top and open bottom, and a plurality of cameras positioned within the unit **316** that are aimed at the top, mid and lower central regions of the interior of the unit **316**, as disclosed, for example, in U.S. Provisional Patent Application Ser. No. 62/269,640 filed Dec. 18, 2015 and U.S. patent application Ser. No. 15/228,692, filed Aug. 4, 2016, the disclosures of which are hereby incorporated by reference in their entireties. The plurality of cameras take images of an object when it is dropped by the end effector **304** through the unit **316**. The unit **316** may also include lights within the unit **316**, and one or more sensors (e.g., laser sensors) at the top of the unit **316** that detect when an object is dropped into the unit **316**. The plurality of cameras are designed to collect a plurality of images of each object from multiple views (ideally all possible views) to aid in identifying or confirming the identity of the dropped object.

[0046] The dropped object then falls into a first carriage **318** that is provided on a track **320** on which the conveyor **318** may be moved automatically between a first sortation stage **322** and a second sortation stage **324** on either side of the area in which the object was dropped. The first carriage **318** is also provided with actuators that may selectively cause the carriage to tip on either side of the track **320** to dump its contents into either the carriage **50** at sortation stage **322** or sortation stage **324**, similar to the operation of the carriage **50** discussed above with reference to FIGS. 1-7. The first sortation stage **322** includes a carriage **50** that may receive objects from the carriage **318**, and which travels along a track between two rows of collection bins into which objects may be dumped along guide walls **240**, and the second sortation stage **324** includes a carriage **50** that may receive objects from the carriage **318**, and which travels

along a track between two rows of collection bins into which objects may be dumped along guide walls **240**.

[0047] The system of FIG. **8** shows a system with two shuttle sorters. When an object is picked from the infeed conveyor, it is dropped onto the first shuttle sorter **318**. That shuttle sorter carries the object to one of two shuttle systems **200**, drops the object in the carrier for that system, and then moves back to home. Each of the carriers for systems **200** may also include a carriage guide **430** (as shown in FIG. **9**) that guides objects into a carriage **50** but does not move with the carriage. Due to the limited travel, this back and forth operation may be performed in the time it takes the articulated arm to pick another object (assuming the articulated arm is picking objects at approximately a human rate of throughput).

[0048] The shuttle sorter system therefore includes an object carriage on a motorized linear slide that travels above a double row of collection bins. The carriage is loaded with an object and then moves along the linear slide until it has reached the collection bin where the object belongs; it then uses rotational actuation to eject the object to one side or the other, where it falls into one of the two collection bins at that location. The carrier then returns to the home position to await another object.

[0049] In the concept as shown, each system **200** is limited to **8** collection bins long, for **16** total collection bins per wing. The length of collection bins traveled by the linear carriage should be balanced with other throughput factors in the system. Given achievable speeds for belt driven linear actuators, distances, and picking speed of the articulated arm, this length of **8** collection bins is a reasonable length that does not adversely limit system throughput (i.e., the articulated arm does not have to wait for a carriage to return to home before picking another object). At this 8x2 or 16 collection bin count, each system **200** has a divert cost far less per intelligent divert for currently fielded solutions, as discussed above.

[0050] Systems in the prior art also do not use back and forth style sortation because the shuttle can only handle one item at a time, and the shuttle needs to return to its home position after each sort. In this system, this concern is alleviated in three ways: 1) multiple systems **200** are used in parallel, 2) frequent destinations are assigned to collection bins closer to the shuttle's home position, thereby reducing the average cycle time of the shuttle, and 3) mapping of objects to collection bins is dynamic and under the control of the system as discussed above with reference to the system of FIGS. **1-7**.

[0051] Systems of the invention are therefore, highly scalable. FIG. **9**, for example, shows a system **400** that includes four shuttle systems **200** of FIG. **5**, each of which includes a carriage **50** on a track **18** that provides objects to bins **242**. The system **400** also includes an articulated arm **402** with an end effector **404**, an input area **406** in which objects are presented for sortation, a primary camera **408** for identifying objects to be sorted, and a receiving conveyor **410** for receiving objects to be sorted from any of a human worker, another conveyor, or an input pan. The system also includes a non-sortable output chute **412** that leads to a non-sortable output bin **414** for providing objects that the system either could not identify or could not sort for any other reason (e.g., could not grasp or pick up).

[0052] Again, in addition to the overhead camera **408**, the system also includes a drop scanner unit **416**, which includes

an open top and open bottom, and a plurality of cameras positioned within the unit **416** that are aimed at the top, mid and lower central regions of the interior of the unit **416**, as discussed above with reference to drop scanner unit **316** of FIG. **8**. The dropped object then falls into a carriage **418** that is provided on a track **420** on which the conveyor **418** may be moved automatically between a first sortation stage **422**, a second sortation stage **424**, a third sortation stage **426**, and a fourth sortation stage **428**. Again, the carriage **418** is also provided with actuators (e.g., electric or pneumatic) that may selectively cause the carriage to tip on either side of the track **420** to dump its contents into the carriage **50** at any of sortation stages **422**, **424**, **426** or **428**, similar to the operation of the carriage **50** discussed above with reference to FIGS. **1-8**.

[0053] The system **400** therefore includes **64** total collection bins. This system may be further scaled to add more collection bins. The first shuttle sorter (that transfers objects from the picking robot to the systems **200**) may also be lengthened to accommodate **4** shuttle systems **200** before system throughput is adversely affected.

[0054] In particular, the system may be further expanded by again doubling the number of systems **200**. This requires the addition of another shuttle sorter that takes the object from the picking robot and delivers it to one of the **4** systems **200**. This keeps the shuttle sort[[]] back and forth travel time from adversely effecting overall system throughput. Such a system is shown in FIG. **10**. FIG. **10** shows a system **500** that includes eight shuttle systems **200** of FIG. **9**, each of which includes a carriage **50** on a track **18** that provides objects to bins **242**. The system **500** also includes an articulated arm **502** with an end effector **504**, an input area **506** in which objects are presented for sortation, and a receiving conveyor **410** for receiving objects to be sorted from any of a human worker, another conveyor, or an input pan. The system also includes a non-sortable output chute **512** that leads to a non-sortable output bin for providing objects that the system either could not identify or could not sort for any other reason (e.g., could not grasp or pick up).

[0055] Again, the system **500** also includes a drop scanner **516**, which includes an open top and open bottom, and a plurality of cameras positioned within the scanner **516** that are aimed at the top, mid and lower central regions of the interior of the scanner **516**, as discussed above with reference to drop scanner **316** of FIG. **10**. The dropped object then falls into one of two carriages **518**, **520** that are provided on a track **522** on which the carriages **518**, **520** may be moved automatically between a first sortation stage **524**, a second sortation stage **526**, a third sortation stage **528**, a fourth sortation stage **530**, a fifth sortation stage **532**, a sixth sortation stage **534**, a seventh sortation stage **536** and an eighth sortation stage **538**. Again, the carriages **518**, **520** are also provided with pneumatic actuators that may selectively cause the carriage to tip on either side of the track **522** to dump its contents into the carriage **50** at any of sortation stages **524-538**, similar to the operation of the carriage **50** discussed above with reference to FIGS. **1-9**. The system **500** therefore includes **64** total collection bins.

[0056] In each of the systems **300**, **400** and **500**, the carriages **318**, **418**, **518**, **520** are able to travel along its track in a direction far enough to reach both the input conveyor as well as the non-sortable output chute. This provides that the system may elect to send an object in the first carriage to

either the input conveyor to be re-processed, or to the non-sortable output chute if the object is not sortable.

[0057] The system also provides, in each embodiment, dynamic collection bin allocation as discussed above. In typical human manned systems, collection bins are statically associated (to destinations, next stop facilities, customers, etc.) and don't change frequently; this is so that efficiency benefits, may be gained by humans learning the association and cubby locations. In the systems discussed above, no such constraints exist, since the system is placing all of the objects in collection bins, and it always has comprehensive knowledge of which objects are in the system, which are in each bin, etc. The systems also have knowledge of all historical sortation activity, meaning that historical trends can be used to make even smarter choices about collection bin allocation.

[0058] If, for example, the historical data suggests that two of the collection bins in this system get the most objects in each sort cycle, then the system will allocate one of these bins to the first system **200** (wing), and one to the second, thus ensuring that all the high volume bins are not on one wing creating a bottleneck. The system may also allocate bins close to the beginning of the wing, thereby ensuring minimum cycle times for the busiest collection bins. Further, if the system needs an empty bin, it can signal to a human operator to come and empty a given bin, allowing that bin to be used as soon as it is emptied. These strategies ensure that the cycle time of the shuttle sort wings does not impact overall system throughput.

[0059] Additionally, the system may also allocate and group objects so as to maximize any other arbitrary cost function. Such a sortation system is almost always a small part of a large system, usually extending across multiple facilities around the state, country, or world. As a part of such a large network, the performance of this system inevitably has impacts on costs elsewhere in the network. By understanding these impacts, the system presented here can allocate objects to collection bins in order to minimize cost impact elsewhere in the macro network.

[0060] In this system concept, additional articulated arms (robots) may also be added to each of the concepts to scale throughput for the system. By adding robots and shuttle sort wings, and tuning shuttle sorter speeds and robot picking/scanning speeds, a wide range of overall system throughputs and collection bin counts are possible using the same basic architecture.

[0061] For further scaling **8** wings fed by one pick/scan station may be a maximum for certain applications. To scale a maximum number of bins and a maximum throughput beyond this, multiple of these stations can be parallelized and fed by manual or automated means, just as manual sort cells are fed in concepts discussed in the prior art. This allows for continued linear scaling of throughput, as well as for greater numbers of collection bins, since the system can now dynamically allocate between all the bins in all the wings in all of the parallel cells.

[0062] Those skilled in the art will appreciate that numerous modifications and variations may be made to the above disclosed embodiments without departing from the spirit and scope of the present invention.

What is claimed is:

1.-23. (canceled)

24. An object processing system comprising:

a carriage including a load bed on which an object is received, and a carrier that supports the load bed;

a track system along which the carrier of the carriage is adapted to reciprocally move to and from locations between first and second opposite ends of the track system, the carriage being movably mounted at a carriage elevation on the track system for movement of the carriage at the carriage elevation, and the track system including a loading location at which the object is loaded onto the load bed of the carriage;

a plurality of destination locations, each destination location including a destination container with an open top, the plurality of destination locations being provided in an array of destination locations arranged under the track system at a destination location elevation that is lower than the carriage elevation; and

an actuation system for causing the carriage to discharge the selected object from the load bed and drop the object into a selected destination container of a selected destination location of the plurality of destination locations, wherein the carriage is driven in a first route to move along the track system from the loading location towards the selected destination location and is driven in a second route that is different than the first route to move along the track system towards the loading location from the selected destination location to receive a next selected object at the loading location.

25. The object processing system of claim **24**, wherein the object processing system further includes an input conveyor on which the plurality of objects are provided to a programmable motion device.

26. The object processing system of claim **25**, wherein the programmable motion device includes an end-effector with a vacuum cup for grasping the selected object.

27. The object processing system of claim **26**, wherein the selected object is dropped from the vacuum cup into the load bed of the carriage.

28. The object processing system of claim **24**, wherein the object processing system further includes a perception system for providing perception data regarding an object.

29. The object processing system of claim **28**, wherein the object processing system further includes a computer processing system for assigning the selected destination location of the plurality of destination locations for the object based on the perception data.

30. The object processing system of claim **24**, wherein the actuation system is actuatable to drop the selected object from the load bed into the selected destination container in a direction that is transverse to a direction of movement of the carrier.

31. The object processing system of claim **24**, wherein the selected destination container is provided on a pull-out drawer.

32. The object processing system of claim **31**, wherein the pull-out drawer includes at least one further destination container associated with a further destination location of the plurality of destination locations.

33. An object processing system comprising:

a perception system for providing perception data regarding an object;

a processing system for assigning a selected destination location of a plurality of destination locations for the object based on the perception data, each of the plurality of destination locations including a destination container with an open top;

a carriage including a load bed on which an object is received, and a carrier that supports the load bed;

a track system along which the carrier of the carriage is adapted to move to and from locations along the track system, the carriage being movably mounted at a carriage elevation on the track system for movement of the carriage at the carriage elevation, and the track system including a loading location at which the object is loaded onto the load bed of the carriage, the plurality of destination locations being provided in an array of destination locations arranged under the track system at a destination location elevation that is lower than the carriage elevation; and

an actuation system for causing the carriage to discharge the selected object from the load bed and drop the object into a selected destination container of a selected destination location of the plurality of destination locations, wherein the carriage is driven in a first route to move along the track system from the loading location towards the selected destination location and is driven in a second route that is different than the first route to move along the track system towards the loading location from the selected destination location to receive a next selected object at the loading location.

34. The object processing system of claim **33**, wherein the object processing system further includes an input conveyor on which the plurality of objects are provided to a programmable motion device.

35. The object processing system of claim **34**, wherein the programmable motion device includes an end-effector with a vacuum cup for grasping the selected object.

36. The object processing system of claim **35**, wherein the selected object is dropped from the vacuum cup into the load bed of the carriage.

37. The object processing system of claim **33**, wherein the actuation system is actuatable to drop the selected object from the load bed into the selected destination container in a direction that is transverse to a direction of movement of the carrier.

38. The object processing system of claim **33**, wherein each of the plurality of destination locations includes an associated chute wall for guiding the object to an associated destination container.

39. The object processing system of claim **33**, wherein the selected destination container is provided on a pull-out drawer.

40. The object processing system of claim **39**, wherein the pull-out drawer includes at least one further destination container associated with a further destination location of the plurality of destination locations.

41. A method of processing objects comprising:

providing perception data regarding an object;

assigning a selected destination location of a plurality of destination locations for the object based on the perception data;

receiving the object on a carriage including a load bed and a carrier that supports the load bed;

moving the carrier of the carriage along a track system along which the carrier of the carriage is adapted to move to and from locations, the carriage being movably mounted at a carriage elevation on the track system for movement of the carriage at the carriage elevation, and the track system including a loading location at which the object is loaded onto the load bed of the carriage;

moving the carrier to the selected destination location of the plurality of destination locations, each destination location including a destination container with an open top, the plurality of destination locations being provided in an array of destination locations arranged under the track system at a destination location elevation that is lower than the carriage elevation; and

causing the carriage to discharge the selected object from the load bed and drop the object into a selected destination container of a selected destination location of the plurality of destination locations, wherein the carriage is driven in a first route to move along the track system from the loading location towards the selected destination location and is driven in a second route that is different than the first route to move along the track system towards the loading location from the selected destination location to receive a next selected object at the loading location.

42. The method of claim **41**, wherein the method further includes providing a plurality of objects using an input conveyor to a programmable motion device.

43. The method of claim **42**, wherein the programmable motion device includes an end-effector with a vacuum cup for grasping the selected object.

44. The method of claim **43**, wherein the selected object is dropped from the vacuum cup into the load bed of the carriage.

45. The method of claim **41**, wherein causing the carriage to discharge the selected object from the load bed and drop the object into the selected destination container involves dropping the selected object from the load bed into the selected destination container in a direction that is transverse to a direction of movement of the carrier.

46. The method of claim **41**, wherein each of the plurality of destination locations includes an associated chute wall for guiding the object to an associated destination container.

47. The method of claim **41**, wherein the selected destination container is provided on a pull-out drawer.

48. The method of claim **47**, wherein the pull-out drawer includes at least one further destination container associated with a further destination location of the plurality of destination locations.

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